INVOICEAI

INVOICE REGION DETECTION & BUSINESS-PARAMETER EXTRACTION FOR OBLIGATION-READINESS

Group 2 – Full Stack

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AGENDA

- PROBLEM STATEMENT
- WHY THIS PROBLEM
- LITERATURE REVIEW
- METHODOLOGY– ML Canvas
- MODEL BENCHMARKING
- MODEL DEPLOYMENT
- DEMO
- Q&A

PROBLEM STATEMENT

Manual invoice review blocks automation

Invoices still arrive as PDFs, scans, screenshots, and email attachments. Before automation starts, people must manually validate what is missing.

Messy input
PDFs · scans · images · emails

→

Manual review
references · totals · terms · approvals

→

Business friction
delays · rework · disputes

→

Late obligation creation

The bottleneck is not reading text — it is validating whether the invoice contains the commercial evidence needed to enter a finance workflow.

Missing

PO / order / contract references

Unclear

payment terms and due dates

Manual

stamp and signature checks

Delayed

AP/AR and dispute workflows

WHY THIS PROBLEM

The intake layer is the missing piece

Pistac.io can create more value if it captures invoices at the document stage and converts them into structured obligation data.

Today
Invoice Document → Manual Review → Spreadsheet / ERP Entry

With Invoice Intelligence
Invoice Document → AI Validation → ERP Obligation Record

→

Downstream platform workflows

AP/AR visibility

Payment tracking

Dispute handling

Credit intelligence

Future netting workflows

Strategic opportunity: move InvoiceAI earlier in the finance workflow — from downstream tracking to first-mile invoice capture.

LITERATURE REVIEW

MARKET ANALYSIS

Market analysis

$6.9–8.0B

Global AP-automation software market, 2026

12–14%

CAGR through the early 2030s

Tool	Positioning
SAP Ariba / Coupa / Basware	Enterprise procure-to-pay; heavy implementation
Tipalti	Mid-market to enterprise; global payouts
Bill.com / Stampli	SMB to mid-market workflow platforms
DL2 InvoiceAI (us)	Transparent, configurable obligation-readiness triage

Sources: Mordor Intelligence 2026; Spherical Insights 2025–2035.

We're not chasing a first-mover advantage — we're claiming the segment the incumbents structurally can't serve, with a verdict engine transparent enough for a compliance-adjacent use case.

InvoiceAI — Machine Learning Canvas

Obligation-readiness triage for Accounts Payable · scoping & planning view

Value proposition · Predict · Learn · Monitoring

PREDICTION TASK

- Multi-stage: object detection (bbox + class per region, 14 region classes + stamp/signature) → OCR on detected crops → rule-based verdict.
- Entity: one invoice image. Outcome: Ready / Needs review / Not ready, observed.

DECISIONS

- Verdict routes the invoice: auto-accept as a structured obligation record, or flag for human review naming the missing field(s).
- Policy rules (visual mark, reference no., date range, payment terms) are user-configurable and fail-closed.

VALUE PROPOSITION

- For back-office/AP teams and platforms like Pistac.io turning scanned invoices into structured obligation records.
- Replaces manual, inconsistent per-invoice checks for stamps, signatures, reference numbers and terms.
- Now live as a callable FastAPI service (port 8000) plus a lightweight browser front-end — not just a demo UI.

DATA COLLECTION

- Initial set sourced from Kaggle invoice/OCR datasets plus a custom hand-annotated subset (50–150 imgs) for labels absent from public data.
- At inference, the raw input is a client-uploaded invoice image; nothing is retained beyond the request.

DATA SOURCES

- Kaggle: High-Quality Invoice Images for OCR; OCR Dataset of Multi-type Documents.
- Kaggle: SignverOD (signatures), StaVer (stamps) — kept as two independent detection classes.

IMPACT SIMULATION

- Cost of a missed field: an invoice wrongly marked Ready skips human review — the fail-closed default is the mitigation.
- Deployment criteria: /health confirms both detector weight sets loaded before traffic is trusted.

MAKING PREDICTIONS

- Real-time, one invoice at a time, via POST /predict (multipart upload).
- ~15–20s per invoice, CPU-only (no GPU in production) on a GCP e2-medium VM.

BUILDING MODELS

- 3 models in production: region detector (YOLOv8n, 14 classes), stamp/signature detector (YOLOv8n, 2 classes), EasyOCR (off-the-shelf).
- SSD was taught; YOLOv8n is the trained artifact — a documented substitution given the project timeline.

FEATURES

- Preprocessed image pixels (grayscale, resize, denoise, deskew) feed the detectors directly — no hand-crafted features.
- OCR text is parsed with keyword/regex matching for 6 reference-number types.

MONITORING

- Live-verified: /health and /policies return correct, schema-matching JSON; a blank test invoice correctly triggers fail-closed "Not ready" on every policy preset.
- docker logs on the VM is the primary signal today — no automated alerting yet, a stated limitation for this graded deployment.
- Known issue under review: the image bundles unused CUDA libraries despite CPU-only inference — caused a real disk-space failure, resolved by resizing 30GB → 100GB.

MODEL BENCHMARKING

Component	Model / Method	Metric	Result	Evaluated On
Stamp Detection	YOLOv8n (2-class)	Precision / Recall / IoU	0.903 / 0.875 / 0.822	SignverOD + StaVer (held-out)
Signature Detection	YOLOv8n (2-class)	Precision / Recall / IoU	0.894 / 0.638 / 0.815	SignverOD + StaVer (held-out)
Region Detection	YOLOv8n (5-class)	Macro Mean IoU	0.873	OCR Dataset test split (98 imgs)
OCR — hard case	EasyOCR	Character Accuracy	78.5% (CER 0.215)	Receipt test split (98 docs)
OCR — easy case	EasyOCR	Character Accuracy	99.98% (CER 0.0002)	Real invoices (120 docs)

Final pipeline output — readiness scoring verdict distribution (750 invoices, mean score 76.3/100)

42% Ready

12.4% Needs Review

45.6% Not Ready

Detection models are strong on held-out data; the honest gap (0/750 stamp/signature on real invoices) is a corpus property, not a model failure — addressed in our implementation plan.

Model Experimentations

Stamp & Signature Detection — 3 Experimental Runs

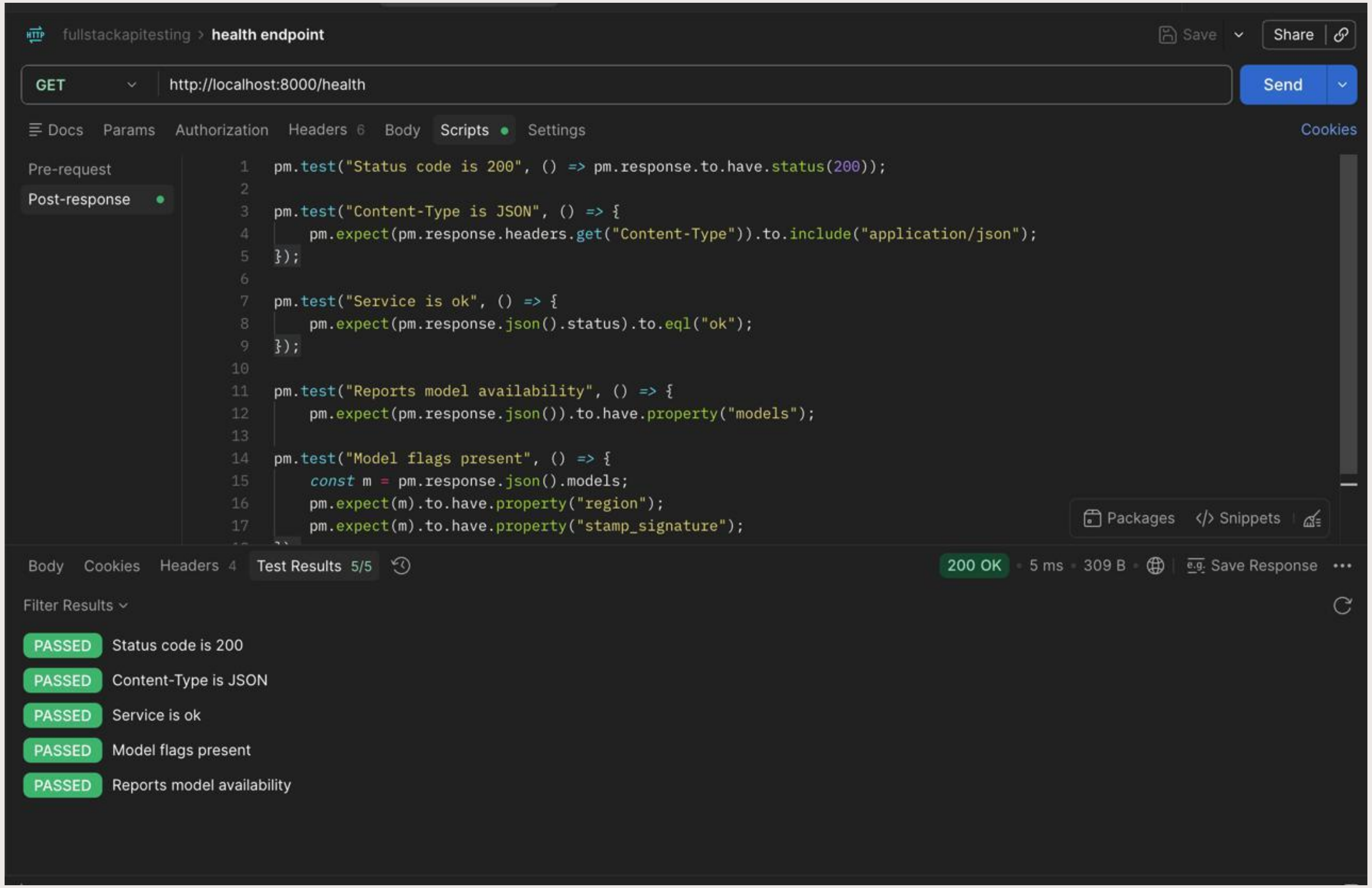
YOLOv8n · single 2-class model (stamp, signature) · per-class metrics on the real held-out split @ conf 0.25

Run	Config (imgsz / epochs / augmentation)	Stamp IoU	Signature IoU	Stamp P / R	Signature P / R	Outcome
Baseline	640 / 50 / default	0.82	0.81	0.903 / 0.894	0.894 / 0.638	Starting point
Attempt 2	768 / 75 / + rotation & shear	0.803	0.804	0.857 / 0.844	0.854 / 0.646	Regressed — aug hurt
Final ✓	**768 / 75 / no rotation·shear**	**0.815**	**0.819**	**0.906 / 0.906**	**0.897 / 0.636**	**Best — kept**

Takeaway: Higher resolution helped only after the harmful rotation/shear was removed — Attempt 2 regressed. IoU sits at the data ceiling (~0.82) across all runs, so the Final config was kept. Signature recall (~0.64) is data-limited (small/faint signatures), not a threshold issue — a confidence sweep confirmed it.

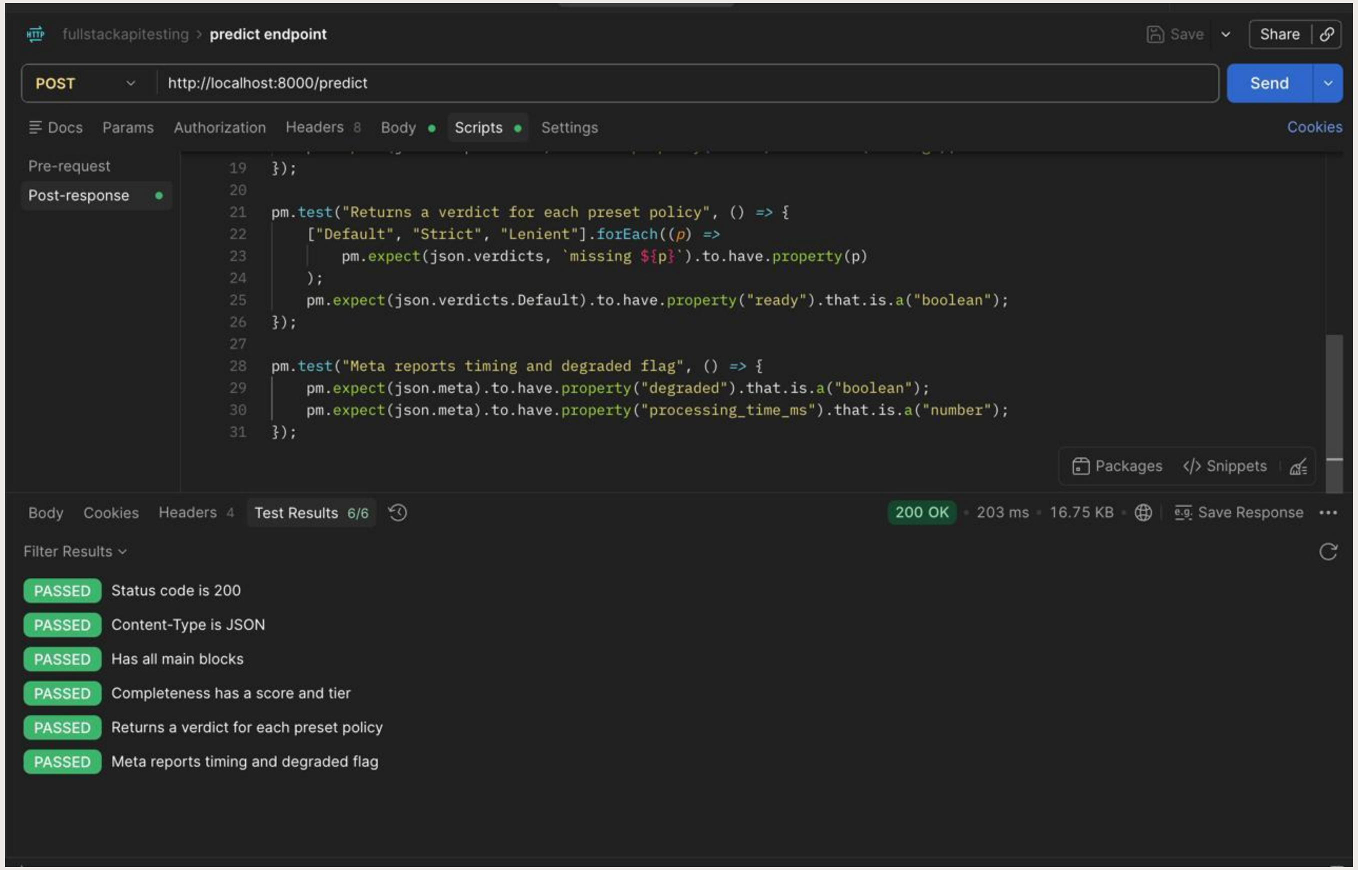
POSTMAN TESTING

- Endpoint predict



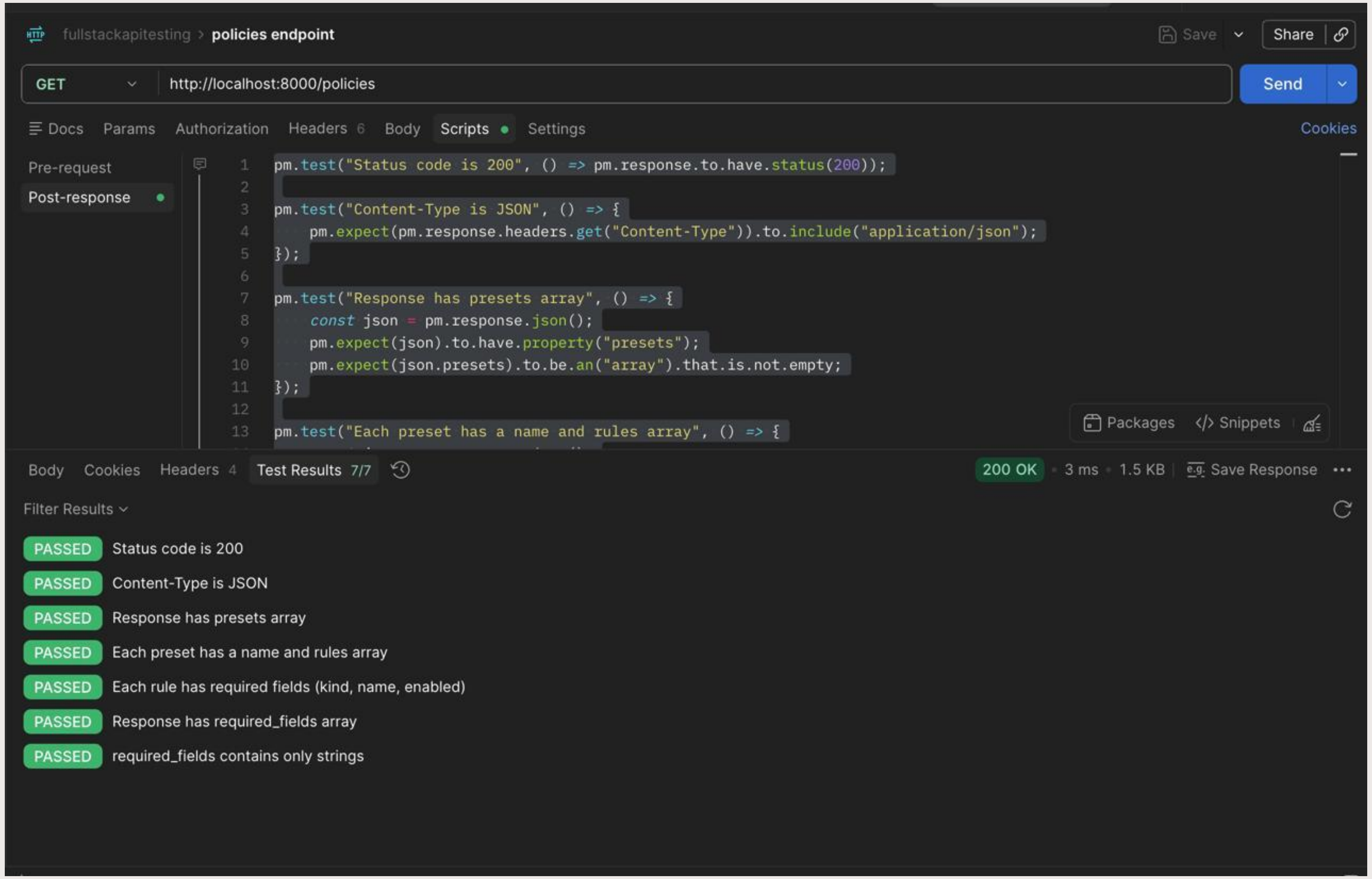
POSTMAN TESTING

- Endpoint predict



POSTMAN TESTING

- Endpoint policies



DOCKERIZATION

Dockerfile U

Dockerfile > ...

```dockerfile
FROM python:3.12-slim

RUN apt-get update && apt-get install -y --no-install-recommends libgl1 libglib2.0-0 curl && rm -rf /var/lib/apt/lists/*

WORKDIR /app

COPY requirements.txt requirements-api.txt ./
RUN pip install --no-cache-dir -r requirements.txt -r requirements-api.txt

COPY . .

EXPOSE 8000

HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
    CMD curl -f http://localhost:8000/health || exit 1

CMD ["uvicorn", "app.api:app", "--host", "0.0.0.0", "--port", "8000"]
```

DEPLOYMENT VALIDATION

```
Problems   Output   Debug Console   Terminal   Ports

PS C:\Users\damir\Full stack project\rolvr-main> docker run -d -p 8000:8000 --name invoiceai-api adzone75/invoiceai-api
Unable to find image 'adzone75/invoiceai-api:latest' locally
latest: Pulling from adzone75/invoiceai-api
44136fa355b3: Already exists
71279a31aa7c: Pull complete
dc534da1c016: Download complete
Digest: sha256:0d26b170aedd7fc82c95fd165135b0b75c7503980ffcc54b0554ad085c82837a
Status: Downloaded newer image for adzone75/invoiceai-api:latest
62e978746d3d4d43eaabd7dea2338fc2cb6215fe3592bb29f427c85ecdd06114
PS C:\Users\damir\Full stack project\rolvr-main> curl.exe -i http://localhost:8000/health
HTTP/1.1 200 OK
date: Fri, 14 Aug 2026 21:46:19 GMT
server: uvicorn
content-length: 183
content-type: application/json

{"status":"ok","service":"invoice-obligation-readiness-api","version":"1.0.0","models":{"region":true,"stamp_signature":true},"ocr_engin
e":"easyocr (CPU, lazy-loaded)","device":"cpu"}
PS C:\Users\damir\Full stack project\rolvr-main> curl.exe -i http://localhost:8000/policies
HTTP/1.1 200 OK
date: Fri, 14 Aug 2026 21:47:19 GMT
server: uvicorn
content-length: 1405
content-type: application/json

{"presets":[{"name":"Default","rules":[{"kind":"visual","name":"Visual mark","enabled":false,"mode":"either"},{"kind":"reference","name"
:"Reference number","enabled":true,"fields":["PO Reference","Order Number","Contract Number"],"mode":"any"},{"kind":"date","name":"Invoi
ce date range","enabled":true,"start":null,"end":null},{"kind":"payment_terms","name":"Payment terms (days)","enabled":false,"op":">","d
ays":30}]},{"name":"Strict","rules":[{"kind":"visual","name":"Visual mark","enabled":true,"mode":"either"},{"kind":"reference","name":"R
eference number","enabled":true,"fields":["PO Reference","Order Number","Contract Number"],"mode":"any"},{"kind":"date","name":"Invoice
date range","enabled":true,"start":null,"end":null},{"kind":"payment_terms","name":"Payment terms (days)","enabled":false,"op":">","days
":30}]},{"name":"Lenient","rules":[{"kind":"visual","name":"Visual mark","enabled":false,"mode":"either"},{"kind":"reference","name":"Re
ference number","enabled":false,"fields":["PO Reference","Order Number","Contract Number","Work Order No."],"mode":"any"},{"kind":"date"
,"name":"Invoice date range","enabled":true,"start":null,"end":null},{"kind":"payment_terms","name":"Payment terms (days)","enabled":fal
se,"op":">","days":30}]}],"required_fields":["PO Reference","Order Number","Contract Number","Work Order No.","Project Reference","Insur
ance Policy Number","Bill of Lading Number"]}
PS C:\Users\damir\Full stack project\rolvr-main> docker ps --filter name=invoiceai-api
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS                        PORTS
                          NAMES
62e978746d3d   adzone75/invoiceai-api   "uvicorn app.api:app…"   About a minute ago   Up About a minute (healthy)   0.0.0.0:8000->8000/t
cp, [::]:8000->8000/tcp   invoiceai-api
PS C:\Users\damir\Full stack project\rolvr-main>
```

MAKING IT A LIVE PUBLIC API

Deploying InvoiceAI on Google Cloud Platform (GCP)

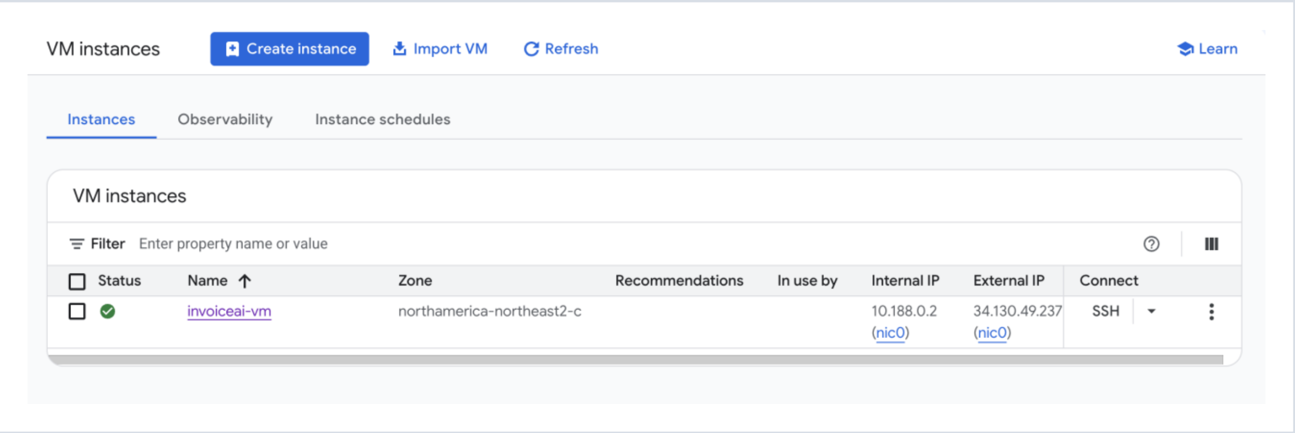
1. **Provisioned the VM**
   GCP project, billing, Compute Engine API enabled.

2. **Opened the port**
   Firewall rule for the container's port, scoped by network tag.

3. **Installed Docker**
   SSH’d in, ran the official install script.

4. **Pulled & ran the image**
   Fixed a mid-pull disk-space failure by resizing 30GB → 100GB.

5. **Verified it live**
   /health and /policies confirmed reachable from outside the VM.



Live now — 34.130.49.237 : 8000 · northamerica-northeast2-c

Region detector · stamp/signature detector · EasyOCR — all loaded and responding over HTTP.

Free-tier friendly: this deployment costs a few cents per hour — and comes down with two commands when the demo is done.

Challenges & Issues Encountered

What went wrong along the way, and how it was resolved

1

✕

Missing real outputs

outputs/metrics & outputs/predictions were gitignored — a fresh clone had no computed numbers.

2

□

Model weights not shipped

models/*/best.pt is gitignored — the API ran “fail-closed” with empty detections until weights were added.

3

!

Disk ran out mid-pull

Image bundled unused CUDA libraries — 30GB boot disk couldn't fit it.

4

↻

First fix didn't take

growpart wasn't installed — the partition never grew, so resize2fs found “nothing to do.”

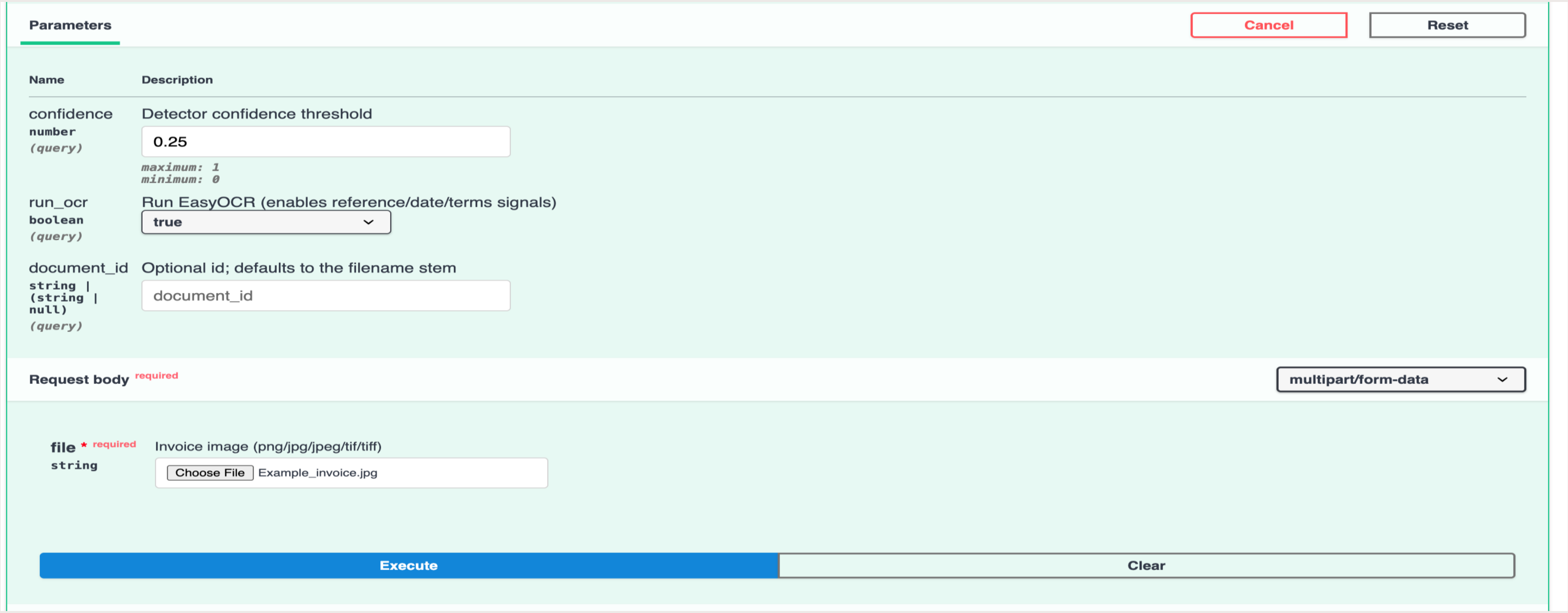
5

?

Wrong app, wrong port

Expected the Streamlit UI (8501) — the deployed image was a JSON-only FastAPI service (8000).

DEMO 1



Parameters
Cancel
Reset
Name
Description
confidence
number
(query)
Detector confidence threshold
0.25
maximum: 1
minimum: 0
run_ocr
boolean
(query)
Run EasyOCR (enables reference/date/terms signals)
true
document_id
string | (string | null)
(query)
Optional id; defaults to the filename stem
document_id
Request body required
multipart/form-data
file * required
string
Invoice image (png/jpg/jpeg/tif/tiff)
Choose File
Example_invoice.jpg
Execute
Clear

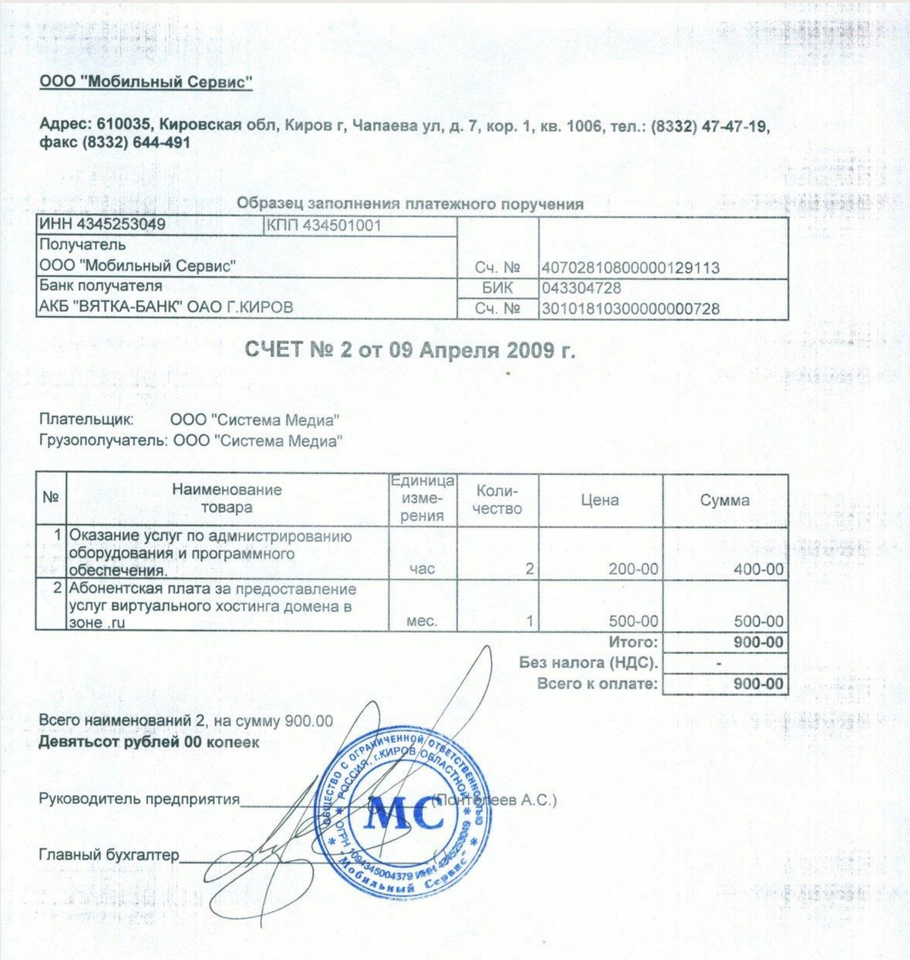
DEMO 1

Response body


```
{
  "document_id": "Example_invoice",
  "filename": "Example_invoice.jpg",
  "image": {
    "width": 1180,
    "height": 1740
  },
  "detections": {
    "regions": [
      {
        "region_label": "other_text",
        "confidence": 0.836,
        "xmin": 751.007568359375,
        "ymin": 636.8477172851562,
        "xmax": 803.6395263671875,
        "ymax": 658.7554931640625
      },
      {
        "region_label": "other_text",
        "confidence": 0.832,
        "xmin": 257.9005432128906,
        "ymin": 649.3091430664062,
        "xmax": 327.8509216308594,
        "ymax": 667.1672973632812
```


Response headers

```
access-control-allow-origin: *
content-length: 12006
content-type: application/json
date: Fri,14 Aug 2026 21:29:55 GMT
server: uvicorn
```



ООО "Мобильный Сервис"

Адрес: 610035, Кировская обл, Киров г, Чапаева ул, д. 7, кор. 1, кв. 1006, тел.: (8332) 47-47-19, факс (8332) 644-491

Образец заполнения платежного поручения

ИНН 4345253049	КПП 434501001		
Получатель ООО "Мобильный Сервис"		Сч. №	40702810800000129113
Банк получателя		БИК	043304728
АКБ "ВЯТКА-БАНК" ОАО Г.КИРОВ		Сч. №	30101810300000000728

СЧЕТ № 2 от 09 Апреля 2009 г.

Плательщик: ООО "Система Медиа"
Грузополучатель: ООО "Система Медиа"

№	Наименование товара	Единица изме-рения	Коли-чество	Цена	Сумма
1	Оказание услуг по адмнистрированию оборудования и программного обеспечения.	час	2	200-00	400-00
2	Абонентская плата за предоставление услуг виртуального хостинга домена в зоне .ru	мес.	1	500-00	500-00
				Итого:	900-00
				Без налога (НДС).	-
				Всего к оплате:	900-00

Всего наименований 2, на сумму 900.00
Девятьсот рублей 00 копеек

Руководитель предприятия__________ (Потолеев А.С.)

Главный бухгалтер__________

DEMO 1

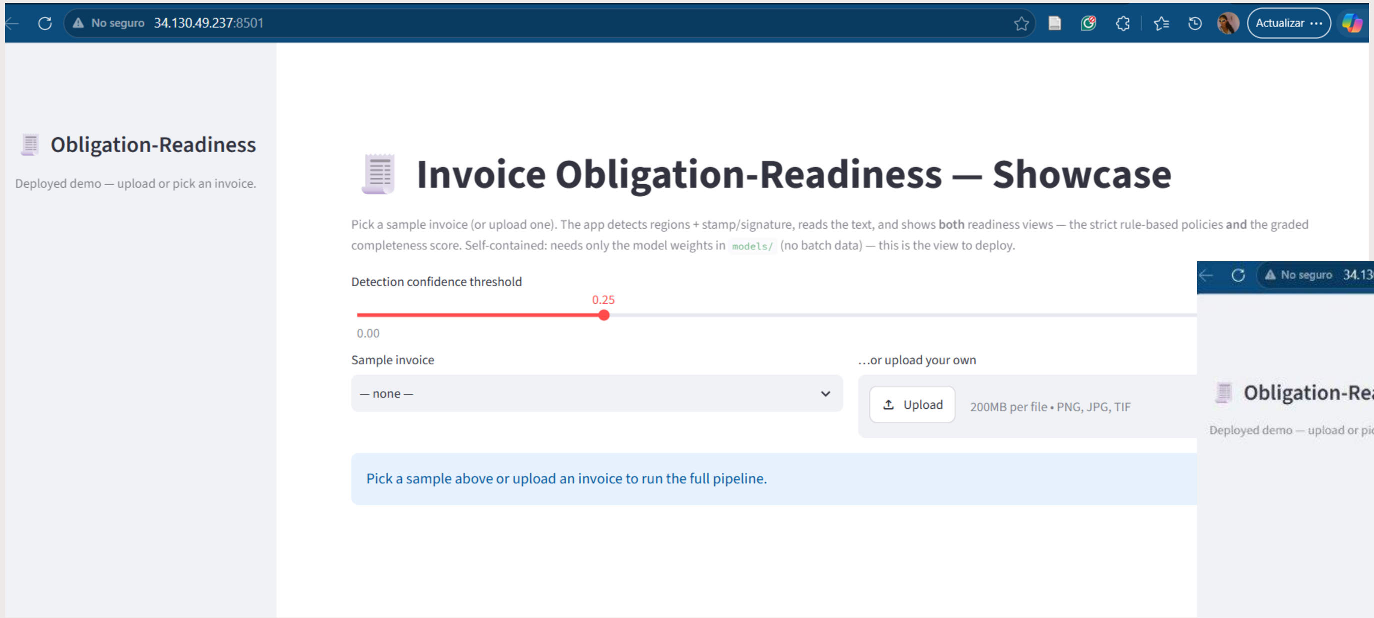
Response body


```
},
"ocr": {
  "ran": true,
  "text": "000 MobunbHbll Cepeuc Anpec: 610035, Kwpobckaa 06n, KwpoB YanaeBa yn, A: 7, Kop pakc (8332) 644-491 1006, Ten : (8332) 47-47-19, OGpa3ey zanonheHwa nnatexhoro nopyye hma knn 434501001 WHH 4345253049 Monyyatenb 000 \" MobwnbHbIM CepBwc' baHk nonyyatena AKs \"BATKA-BAHK\" OAO KMPOB C4; Ng 40702810800000129113 BMK 043304728 C4. No 30101810300000 000728 CYET Ne 2 oT 09 Anpena 2009 r. NnatenbLymk: 000 \"Cuctema Menma' Tpy3ononyyatenb: 000 \"Cucrema Menma' EpMhMLaI MJme Konm- 4eCTBO Pehma Hawmehobahme TOBapa Heha Cymma Okaj ahme ycnyr no anMhMcTpMpoBaHMIQ oGopy4OBaHMA nporparmhoro obecneyehma Abohentckaa nnata 3a npepoctabnehme ycnyr BhpTyanbHoro xocthhra AoMeha 30he 200-00 400-00 Kec 500-00 Mtoro B e3 Hanora (HAC) Bcero onnate: 500-00 900-00 900-00 Bcero Hawmehobahhh Ha cyMmy 900.00 AeBatbcor py6new 00 koneek PykobonMtenb npeAnpMATMA eee AC.) MCc TnabhblM byxrantep_",
  "char_count": 864
},
"signals": {
  "stamp_detected": true,
  "signature_detected": false,
  "references": {
    "PO Reference": false,
    "Order Number": false,
    "Contract Number": false,
    "Work Order No.": false,
    "Project Reference": false,
    "Insurance Policy Number": false,
    "Bill of Lading Number": false
  },
  "invoice_date": "47-47-19",
  "billing_due_days": null
},
```

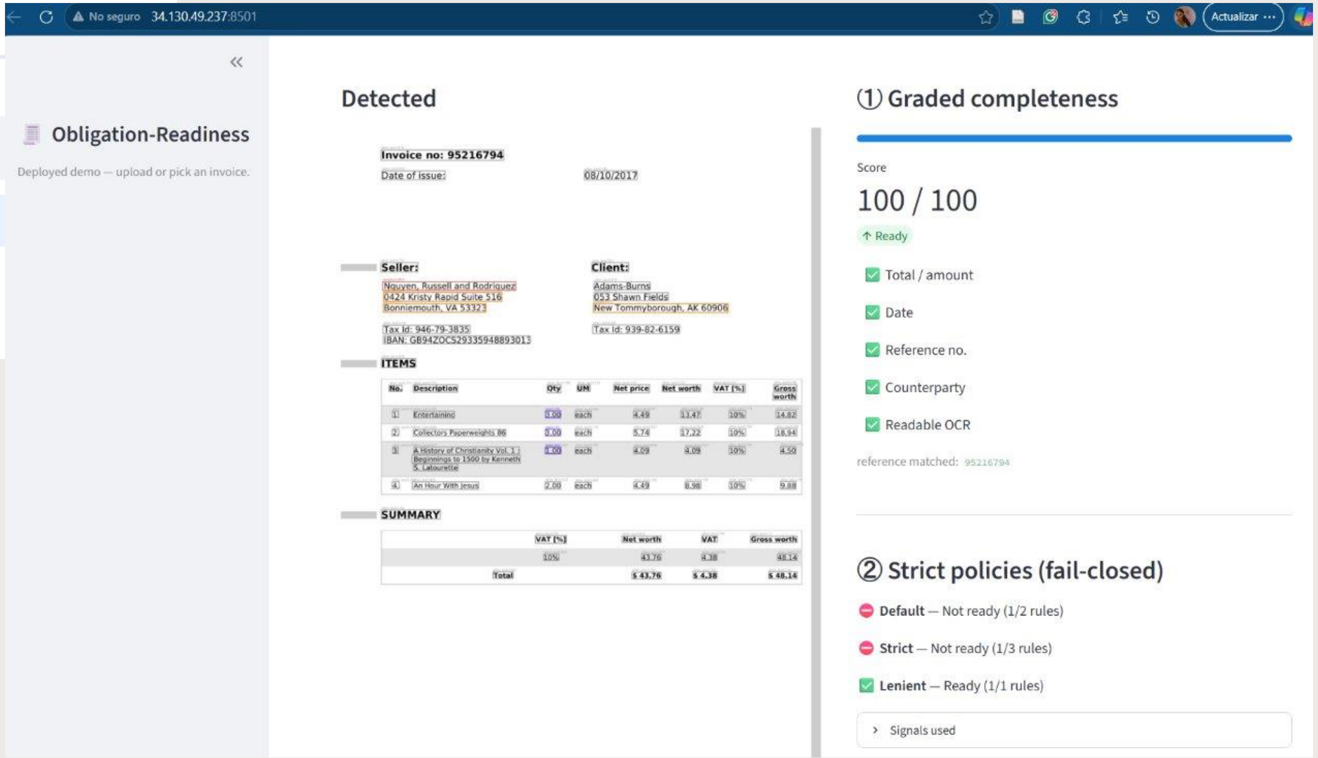

Response body


```
},
"completeness": {
  "score": 45,
  "tier": "Not ready",
  "reference_match": "",
  "has_total": false,
  "has_date": true,
  "has_reference": false,
  "has_counterparty": false,
  "has_readable": true,
  "bonus_payment_terms": false,
  "bonus_visual_mark": true
```

DEMO 2.



No seguro 34.130.49.237:8501
Actualizar
Obligation-Readiness
Deployed demo — upload or pick an invoice.
Invoice Obligation-Readiness — Showcase
Pick a sample invoice (or upload one). The app detects regions + stamp/signature, reads the text, and shows both readiness views — the strict rule-based policies and the graded completeness score. Self-contained: needs only the model weights in models/ (no batch data) — this is the view to deploy.
Detection confidence threshold
0.25
0.00
Sample invoice
— none —
…or upload your own
Upload
200MB per file • PNG, JPG, TIF
Pick a sample above or upload an invoice to run the full pipeline.



No seguro 34.130.49.237:8501
Actualizar
Obligation-Readiness
Deployed demo — upload or pick an invoice.
Detected
Invoice no: 95216794
Date of issue:
08/10/2017
Seller:
Client:
ITEMS
SUMMARY
(1) Graded completeness
Score
100 / 100
Ready
Total / amount
Date
Reference no.
Counterparty
Readable OCR
reference matched: 95216794
(2) Strict policies (fail-closed)
Default — Not ready (1/2 rules)
Strict — Not ready (1/3 rules)
Lenient — Ready (1/1 rules)
Signals used

DEMO 2.

Detection confidence threshold

0.25

Sample invoice

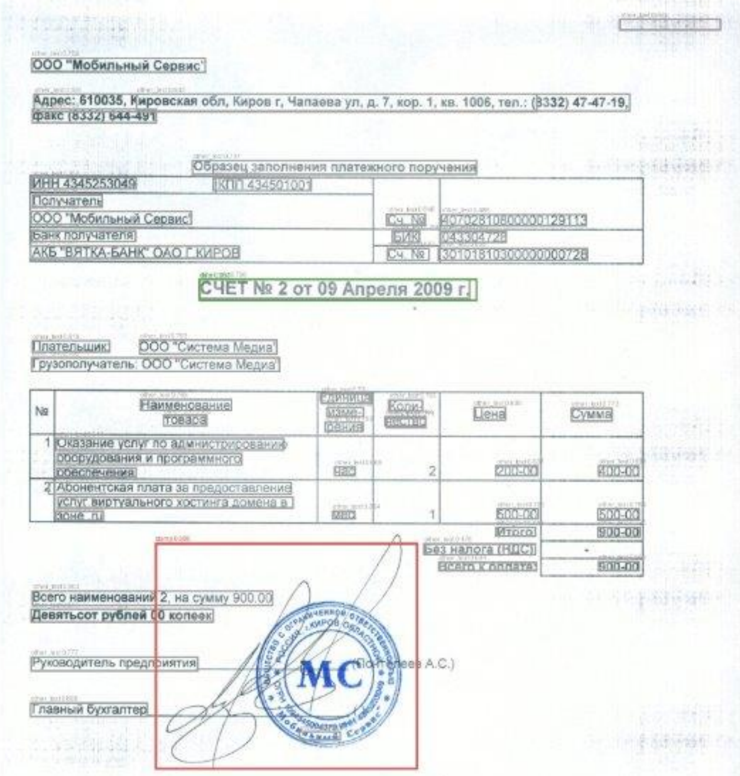
— none —

- both_stamp_and_signature.jpg
- clean_1_invoice.jpg
- clean_2_invoice.jpg
- clean_3_invoice.jpg
- receipt_1_soonhuat.jpg
- receipt_2_abcho.jpg
- signature_only_1_real.png
- signature_only_2_invoice.jpg

Obligation-Readiness

Deployed demo — upload or pick an invoice.

Detected



① Graded completeness

Score

45 / 100

↑ Not ready

- ❌ Total / amount
- ✅ Date
- ❌ Reference no.
- ❌ Counterparty
- ✅ Readable OCR

② Strict policies (fail-closed)

- ⛔ **Default** — Not ready (0/2 rules)
- ⛔ **Strict** — Not ready (1/3 rules)
- ⛔ **Lenient** — Not ready (0/1 rules)

Signals used


```
{
  "stamp_detected" : true
  "signature_detected" : false
  "references" : {
    "PO Reference" : false
    "Order Number" : false
    "Contract Number" : false
    "Work Order No." : false
```

Q & A

THANK YOU!!!

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